



Universal Mobile Access

What is it?

UMA (Universal Mobile Access) is a technology that allows you to use your GSM phone on accessible WiFi, WiMax, and Bluetooth networks. The local network devices become like mini cell sites.

How does it work?

A UMA-enabled mobile has GSM and Wi-Fi radios that can work simultaneously. When the phone moves into the range of a Wireless LAN (connected to the Internet), the user connects to a UMA Network Controller (UNC). The UNC authenticates and authorises the user and updates the IP and MAC addresses on the central network of the mobile service provider. This done, all GSM services are routed through IP to the location of the user. The switching of network happens automatically and seamlessly, without the user's knowledge.

Why will anyone be interested?

UMA reduces the burden on cellular towers, and covers poor-coverage areas such as inside buildings, tunnels, and basements. It reduces transmission costs of calls as well. Customers see fewer call drops, better voice clarity, and increased browser speeds.

Who developed it?

It is a standard accepted by members of the 3rd Generation Partnership Project (3GPP).

Where is it currently deployed?

T-Mobile has been providing the UMA system in the US, and now has 8,500 WiFi hotspots. TeliaSonera, a telecom company, provides the service in Norway, Finland, and Sweden.

When will you see it in India?

We don't know about other manufacturers, but you can expect to see India's first UMA-enabled Nokia phones in the first quarter of 2008.

temperature, and will allow the creation of denser drives that will also last longer.

Back in June 1995, Fujitsu had announced a different method of creating disk drives - by using an anodised aluminium coating. This creates "nano-dots" on the surface of the disk, which are later magnetised to store a single bit of data. The anodised aluminium protects each dot *individually*, thereby reducing the chances of data corruption. (This is known as the patterned media approach, which we've spoken about in an earlier issue.) Currently, the company has announced the radius of each dot to be around 25 nanometres. Fujitsu aims to bring that down to 13 nanometres, which makes for 4 TB of data per square inch! The company hopes to get this technology out by 2010.

But while we're going on and on about hard disk technology, we're wondering what happened to holographic memory! Last we heard, there was a lot of buzz around its capabilities: 4 GB in a volume at the millimetre range.

Daewoo Electronics had also designed a working prototype, and, indeed, TDK came out with holographic-storage disks - but the drive is ridiculously expensive.

In conclusion, it does look like there's plenty to look forward to when it comes to the good old hard disk - it's *not* going to die anytime soon! Unless there's a breakthrough in nanotube storage - but that's a big, big "unless."

AHEAD OF ITS TIME?

Dump Vista, They're Saying

The general consensus in tech circles for some time now has been that Windows Vista does not deliver what it promised, or work how it should, or satisfy. That idea has now been reinforced by



Illustrations Harsho Mohan Chatteraj

the Dutch Consumers Association: it has said all computer outlets in The Netherlands must provide XP "downgrades" to all Vista users who are not satisfied with it.

As you might know, this is not the first time a body has said, in effect, "Perhaps Vista isn't good"; in April, Dell had buckled (in the US) under consumer pressure to give users the option to select between XP and Vista after having made Vista the only operating system available on new laptops and desktops.

The fact that Microsoft has extended the availability of Windows XP

by five months—up until June 2008—also isn't going to help the sales of the new operating system. As if attesting to the tech community's claim that the OS was released before it had been sufficiently tested for bugs, the company offered to give disgruntled Vista customers a copy of Windows XP.

A recent survey by an independent specialist business and consumer technology research

company, Context, indicated that sales of Vista have actually dropped by 17 per cent in the last few months, contrasting with steady sales of Windows XP.

Vista has been plagued by product launch delays—the OS was supposed to have been launched in June 2006, and eventually only became available in January 2007—and poorly-performing drivers (such as those for NVIDIA graphics cards). Bucketsful of bugs will be addressed in Service Pack 1.

Vista critics stand redeemed: Microsoft plans to release Service Pack 3 for Windows XP by early 2008, which will borrow features from Vista!

Microsoft's constant patch releases for the seemingly ill-fated OS has not stopped well-known bloggers like Om Malik from dismissing Vista as being unstable and an unnecessary upgrade.

"Reinstall XP, Remove Vista," goes the header of one of Malik's posts dated May 2007. In fact, who has said Vista is just great? People know it's pretty, and some eminent Netizens have said they'd gotten "very used to it," but that's about it. Still, Service Pack 1 might just change everything, let's not forget. ☒